

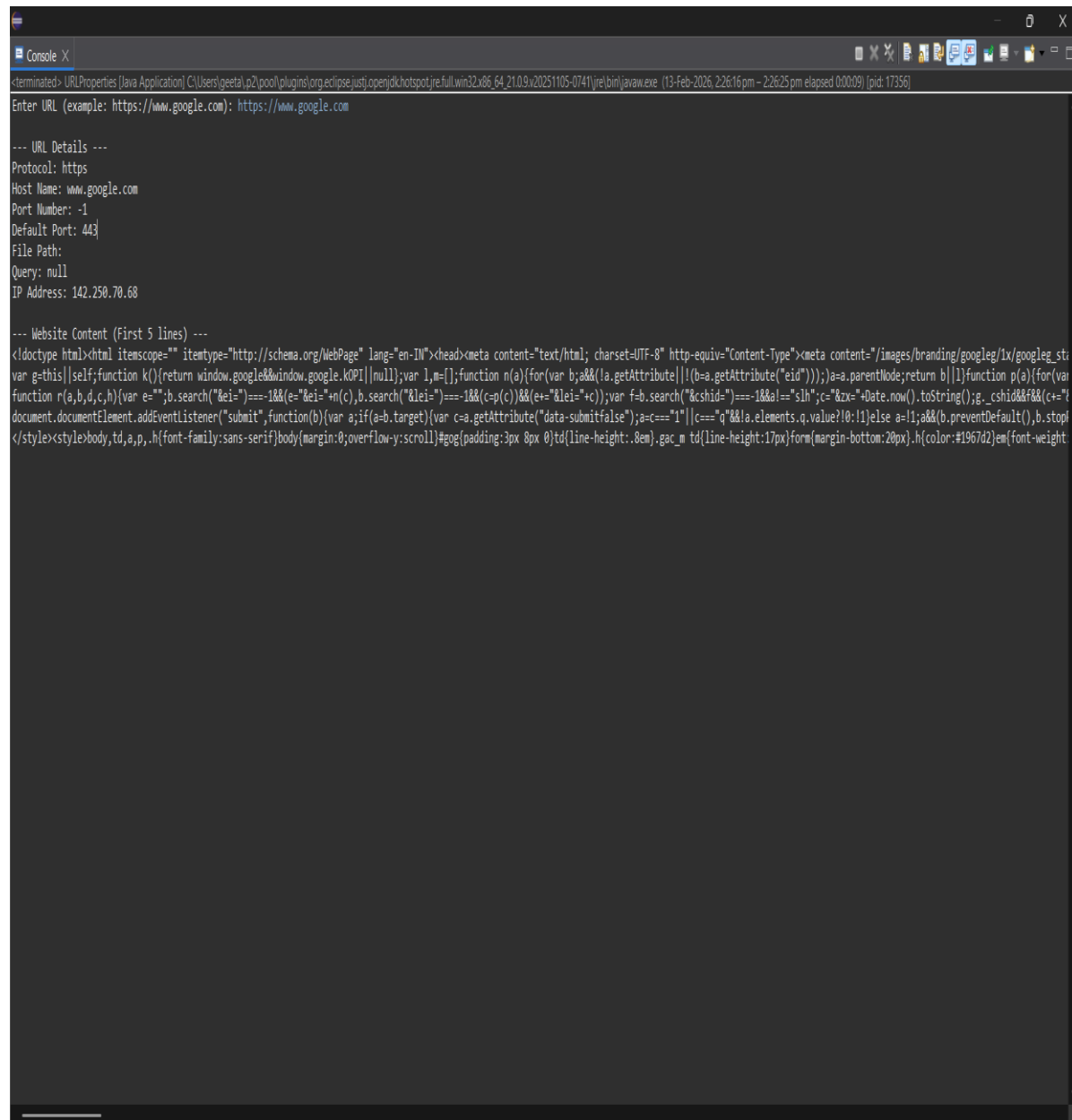
Assignment No	: 4
Roll No	: 381
Batch	: E4
PRN No	: 202502040032
Title of the Assignment	: Networking

- a) Retrieve IP Address and Content of a Given URL
b) Connection-Oriented and Connection-Less Communication Programs

<i>URLProperties.java</i>
<pre>package BasicCodes; import java.net.*; import java.io.*; public class URLProperties { public static void main(String[] args) throws Exception { BufferedReader br = new BufferedReader(new InputStreamReader(System.in)); System.out.print("Enter URL (example: https://www.google.com): "); String urlString = br.readLine(); URL url = new URL(urlString); System.out.println("\n--- URL Details ---"); // Protocol System.out.println("Protocol: " + url.getProtocol()); // Host Name System.out.println("Host Name: " + url.getHost()); // Port Number System.out.println("Port Number: " + url.getPort()); // Default Port System.out.println("Default Port: " + url.getDefaultPort()); // File Path System.out.println("File Path: " + url.getPath()); // Query System.out.println("Query: " + url.getQuery()); // IP Address InetAddress address = InetAddress.getByName(url.getHost()); System.out.println("IP Address: " + address.getHostAddress()); // Display Content (First 5 lines) System.out.println("\n--- Website Content (First 5 lines) ---"); BufferedReader in = new BufferedReader(new InputStreamReader(url.openStream())); for (int i = 0; i < 5; i++) { String line = in.readLine();</pre>

```
        if (line == null) break;
        System.out.println(line);
    }

    in.close();
}
```

Output :A screenshot of a Java application window titled "Console X". The window shows the output of a program that fetches and displays details for the URL "https://www.google.com". The output is divided into two sections: "URL Details" and "Website Content (First 5 lines)". The "URL Details" section lists the protocol (https), host name (www.google.com), port number (-1), default port (443), file path, query (null), and IP address (142.250.70.68). The "Website Content" section displays the first 5 lines of the HTML document, which is a Google search page. The HTML code is partially visible, showing the doctype, meta tags, and the beginning of the body. The application window has a standard Windows-style title bar with minimize, maximize, and close buttons. The console output is displayed in a monospaced font on a dark background.

```
<terminated> URLProperties [Java Application] C:\Users\geeta\p2\pool\plugins\org.eclipse.justopenjdk\hotspot\re.full.win32.x86_64_21.0.9.v20251105-0741\jre\bin\java.exe (13-Feb-2026, 2:26:16 pm - 2:26:25 pm elapsed 0:00:09) [pid: 17356]

Enter URL (example: https://www.google.com): https://www.google.com

--- URL Details ---
Protocol: https
Host Name: www.google.com
Port Number: -1
Default Port: 443
File Path:
Query: null
IP Address: 142.250.70.68

--- Website Content (First 5 lines) ---
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en-IN"><head><meta content="text/html; charset=UTF-8" http-equiv="Content-Type"><meta content="/images/branding/google/1x/google_st
var g=this|self,function k(){return window.google&&window.google.kOPI||null};var l,m=[];function n(a){for(var b;a&&(la.getAttribute("eid")));a=a.parentNode;return b||l}function p(a){for(var
function r(a,b,d,c,h){var e="";b.search("&ei=")===-1&&(e="&ei="+n(c).b.search("&lei=")===-1&&(c-q(c))&&(e+="&lei="+c));var f=b.search("&cshid=")===-1&&a=="slh";c="&zv="+Date.now().toString();g._cshid&&f&&(c+
document.documentElement.addEventListener("submit",function(b){var a;if(a=b.target){var c=a.getAttribute("data-submitfalse");a=c=="1"||c=="q"&&a.elements.q.value?l0:l1}else a=l1;a&&(b.preventDefault()),b.sto
</style><style>body,td,a,p,.h{font-family:sans-serif}body{margin:0;overflow-y:scroll}#gog{padding:3px 8px 0}td{line-height:.8em}.gac_m td{line-height:17px}form{margin-bottom:20px}.h{color:#1967d2}em{font-weight
```

TCPServer.java

```
package TCP;

import java.net.*;
import java.io.*;

public class TCPServer {
    public static void main(String[] args) throws Exception {

        ServerSocket serverSocket = new ServerSocket(5000);
        System.out.println("Server is waiting for connection...");

        Socket socket = serverSocket.accept();
        System.out.println("Client connected!");

        BufferedReader input = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

        PrintWriter output = new PrintWriter(
            socket.getOutputStream(), true);

        BufferedReader br = new BufferedReader(
            new InputStreamReader(System.in));

        while (true) {

            // Receive message
            String message = input.readLine();
            System.out.println("Received: " + message);

            // Take reply using BufferedReader
            System.out.print("Enter reply: ");
            String reply = br.readLine();

            output.println(reply);
        }
    }
}
```

TCPClient.java

```
package TCP;

import java.net.*;
import java.io.*;

public class TCPClient {
    public static void main(String[] args) throws Exception {

        Socket socket = new Socket("localhost", 5000);
        System.out.println("Connected to server!");

        BufferedReader input = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

        PrintWriter output = new PrintWriter(
            socket.getOutputStream(), true);

        BufferedReader br = new BufferedReader(
```

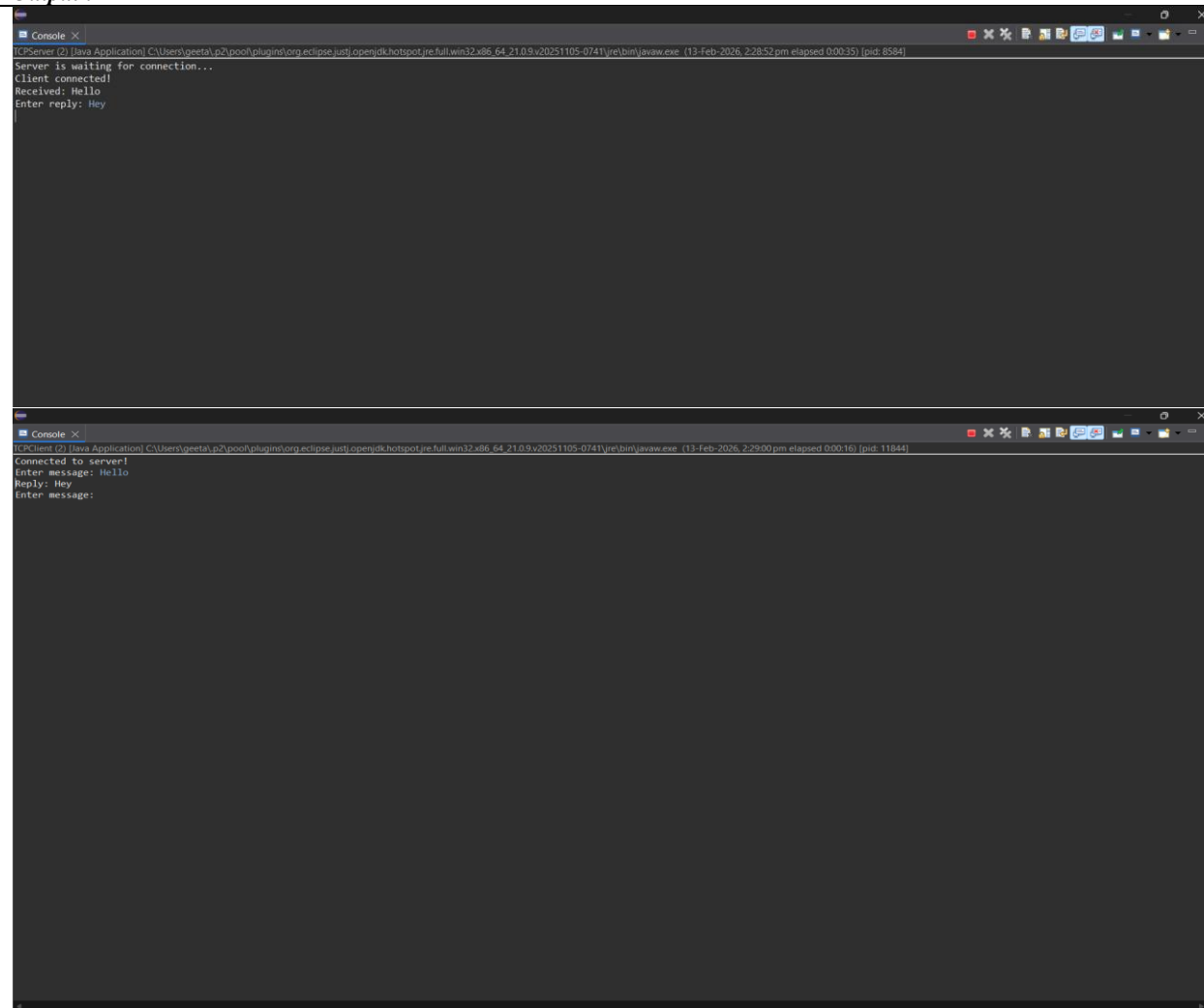
```
new InputStreamReader(System.in));

while (true) {

    // Take input using BufferedReader
    System.out.print("Enter message: ");
    String message = br.readLine();

    output.println(message);

    // Receive reply
    String reply = input.readLine();
    System.out.println("Reply: " + reply);
}
}
```

Output :

The screenshot displays two console windows from the Eclipse IDE. The top window, titled 'Console X', shows the output of a Java server application. It starts with 'Server is waiting for connection...', followed by 'Client connected!', 'Received: Hello', and 'Enter reply: Hey'. The bottom window, also titled 'Console X', shows the output of a Java client application. It starts with 'Connected to server!', followed by 'Enter message: Hello', 'Reply: Hey', and 'Enter message:'. Both windows show the standard Eclipse IDE window title bar and a toolbar with various icons.

```
Console X
[CPServer (2) [Java Application] C:\Users\geeta\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.9.v20251105-0741\jre\bin\javaw.exe (13-Feb-2026, 2:28:52 pm elapsed 0:00:35) [pid: 8584]
Server is waiting for connection...
Client connected!
Received: Hello
Enter reply: Hey

Console X
[CPClient (2) [Java Application] C:\Users\geeta\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.9.v20251105-0741\jre\bin\javaw.exe (13-Feb-2026, 2:29:00 pm elapsed 0:00:16) [pid: 11844]
Connected to server!
Enter message: Hello
Reply: Hey
Enter message:
```

UDPServer.java

```
package UDP;

import java.net.*;
import java.io.*;

public class UDPServer {
    public static void main(String[] args) throws Exception {

        DatagramSocket socket = new DatagramSocket();
        InetAddress ip = InetAddress.getByName("localhost");

        byte[] sendData;
        byte[] receiveData = new byte[1024];

        BufferedReader br = new BufferedReader(
            new InputStreamReader(System.in));

        while (true) {

            // Take input from user using BufferedReader
            System.out.print("Enter message: ");
            String message = br.readLine();

            sendData = message.getBytes();

            DatagramPacket sendPacket =
                new DatagramPacket(sendData, sendData.length, ip, 5000);

            socket.send(sendPacket);

            // Receive reply
            DatagramPacket receivePacket =
                new DatagramPacket(receiveData, receiveData.length);

            socket.receive(receivePacket);

            String reply = new String(receivePacket.getData(), 0,
                receivePacket.getLength());

            System.out.println("Reply: " + reply);
        }
    }
}
```

UDPClient.java

```
package UDP;

import java.net.*;
import java.io.*;

public class UDPClient {
    public static void main(String[] args) throws Exception {

        DatagramSocket socket = new DatagramSocket(5000);
        byte[] receiveData = new byte[1024];
        byte[] sendData;

        BufferedReader br = new BufferedReader(
            new InputStreamReader(System.in));

        System.out.println("Receiver is running...");

        while (true) {

            // Receive message
            DatagramPacket receivePacket =
                new DatagramPacket(receiveData, receiveData.length);

            socket.receive(receivePacket);

            String message = new String(receivePacket.getData(), 0,
                receivePacket.getLength());

            System.out.println("Received: " + message);

            // Take reply input using BufferedReader
            System.out.print("Enter reply: ");
            String reply = br.readLine();

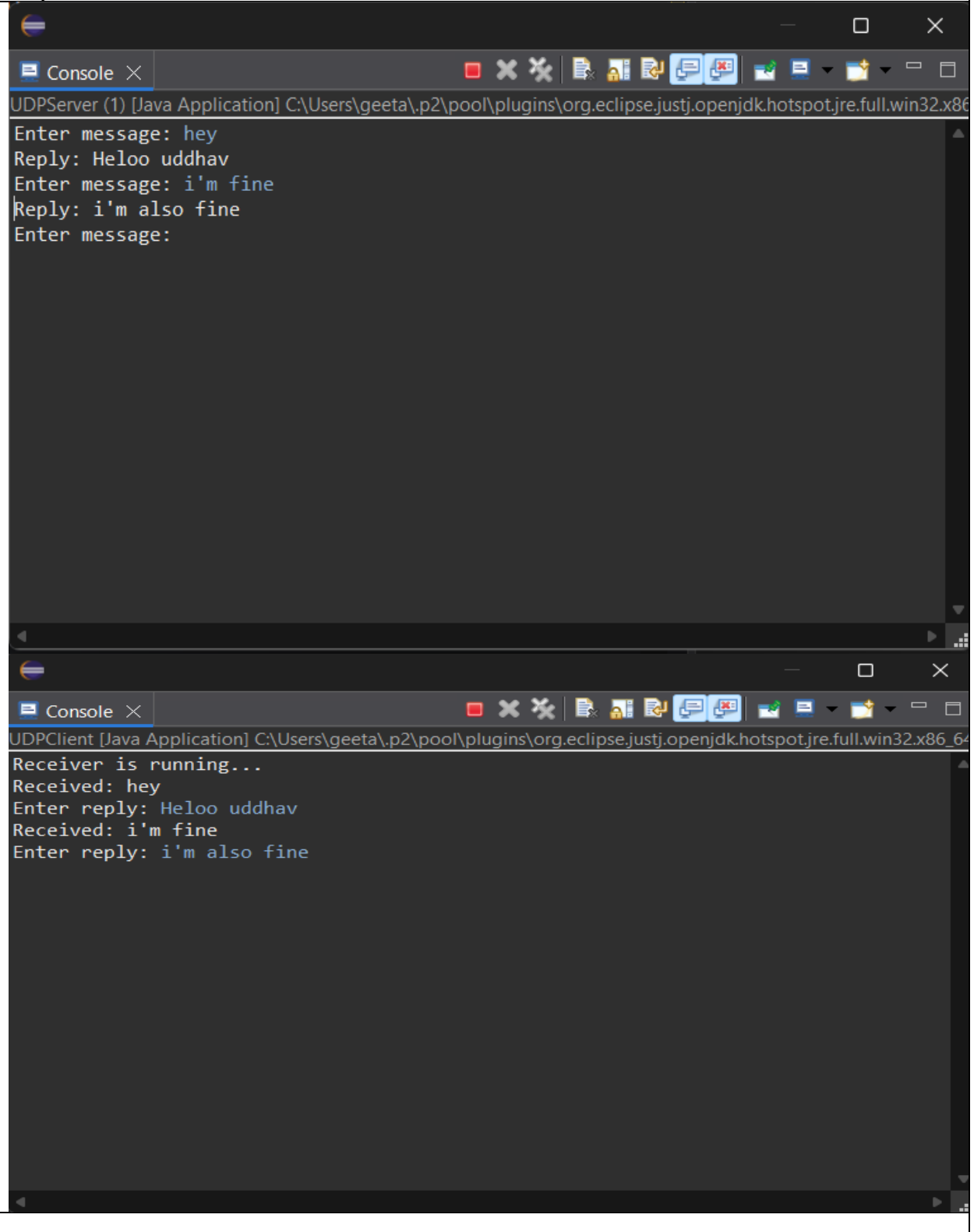
            sendData = reply.getBytes();

            InetAddress senderAddress = receivePacket.getAddress();
            int senderPort = receivePacket.getPort();

            DatagramPacket sendPacket =
                new DatagramPacket(sendData, sendData.length,
                    senderAddress, senderPort);

            socket.send(sendPacket);
        }
    }
}
```

Output :



The image shows two Eclipse IDE console windows. The top window is titled 'UDPServer (1) [Java Application]' and shows the following text: 'Enter message: hey', 'Reply: Heloo uddhav', 'Enter message: i'm fine', 'Reply: i'm also fine', and 'Enter message:'. The bottom window is titled 'UDPClient [Java Application]' and shows the following text: 'Receiver is running...', 'Received: hey', 'Enter reply: Heloo uddhav', 'Received: i'm fine', and 'Enter reply: i'm also fine'. Both windows have a toolbar with icons for running, stopping, and other IDE functions.

```
UDPServer (1) [Java Application] C:\Users\geeta\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.jre\bin\java.exe
Enter message: hey
Reply: Heloo uddhav
Enter message: i'm fine
Reply: i'm also fine
Enter message:

UDPClient [Java Application] C:\Users\geeta\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.jre\bin\java.exe
Receiver is running...
Received: hey
Enter reply: Heloo uddhav
Received: i'm fine
Enter reply: i'm also fine
```

Submitted by,
Uddhav Ramprasad Tour